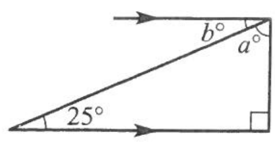
Exercise 3I

Understanding

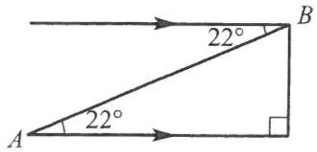
1 Find the values of the pronumerals in this diagram.



2 For this diagram:

a What is the angle of elevation of B from A?

b What is the angle of depression of A from B?



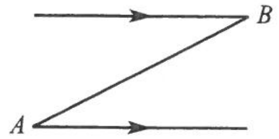
3 Draw this diagram and complete these tasks.

a Mark in the following:

i the angle of elevation (θ) of B from A.

ii the angle of depression (α) of A from B.

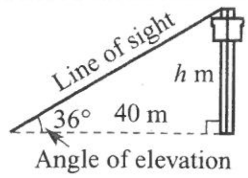
b Is $\theta = \alpha$ in your diagram? Why?



Fluency

Example 15

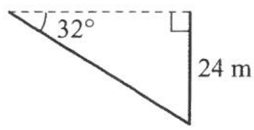
4 The angle of elevation of the top of a tower from a point on the ground 40 m from the base of the tower is 36°. Find the height of the tower to the nearest metre.



5 The angle of elevation of the top of a castle wall from a point on the ground 25 m from the base of the castle wall is 32°. Find the height of the castle wall to the nearest metre.

6 From a point on the ground, Emma measures the angle of elevation of an 80 m tower to be 27°. Find how far Emma is from the base of the tower, correct to the nearest metre.

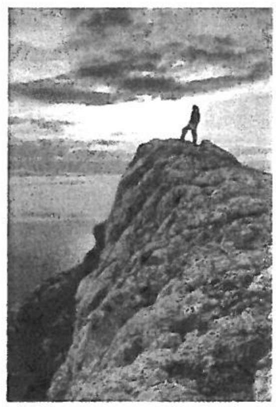
7 From a pedestrian overpass, Chris spots a landmark at an angle of depression of 32°. How far away (to the nearest metre) is the landmark from the base of the 24 m high overpass?



8 From a lookout tower, David spots a bushfire at an angle of depression of 25°. If the lookout tower is 42 m high, how far away (to the nearest metre) is the bushfire from the base of the tower?

Example 16

9 From the top of a vertical cliff, Josh spots a swimmer out at sea. If the top of the cliff is 38 m above sea level and the swimmer is 50 m away from the base of the cliff, find the angle of depression from Josh to the swimmer, to the nearest degree.



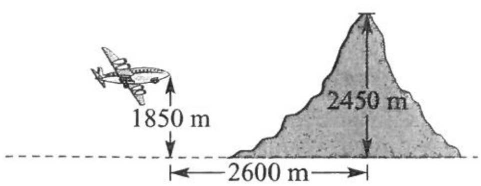
10 From a ship, a person is spotted floating in the sea 200 m away. If the viewing position on the ship is 20 m above sea level, find the angle of depression from the ship to person in the sea. Give your answer to the nearest degree.

11 A power line is stretched from a pole to the top of a house. The house is 4.1 m high and the power pole is 6.2 m high. The horizontal distance between the house and the power pole is 12 m. Find the angle of elevation of the top of the power pole from the top of the house, to the nearest degree.

Problem-solving

Example 17

12 A plane flying at 1850 m starts to climb at an angle of 18° to the horizontal when the pilot sees a mountain peak 2450 m high, 2600 m away from him in a horizontal direction. Will the pilot clear the mountain?



13 A road has a steady gradient of 1 in 10.

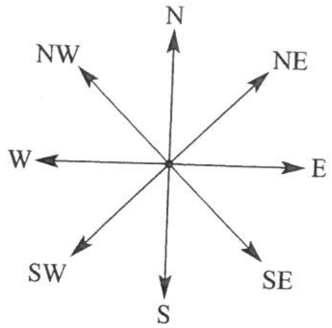
a What angle does the road make with the horizontal? Give your answer to the nearest degree.

b A car starts from the bottom of the inclined road and drives 2 km along the road. How high vertically has the car climbed? Use your rounded answer from part **a** and give your answer correct to the nearest metre.

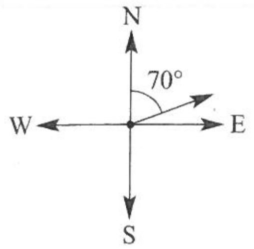
Exercise 3J

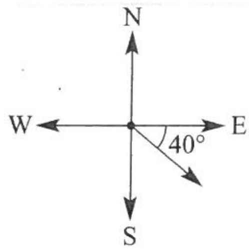
1 Give the true bearings for these common directions.

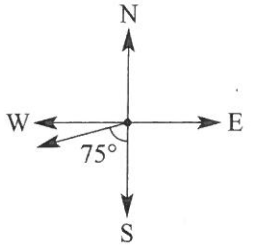
a North (N)
b North-east (NE)
c East (E)
d South-east (SE)
e South (S)
f South-west (SW)
g West (W)
h North-west (NW)

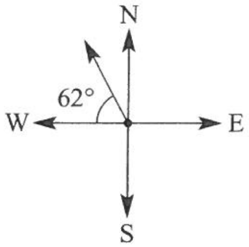


Understanding

a


b


c


d


Understandi

ements.
s to
e for *d*.

O.

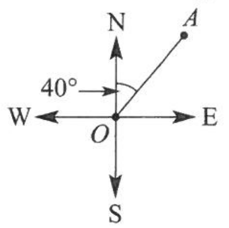
Understanding

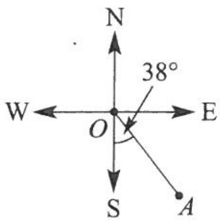
3

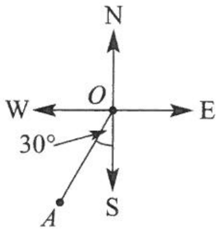
Fluency

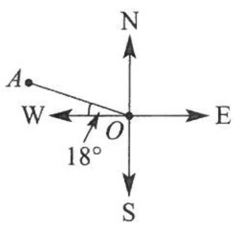
Example 18 3 For each diagram shown, write:

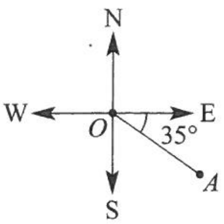
i the true bearing of *A* from *O*
ii the true bearing of *O* from *A*.

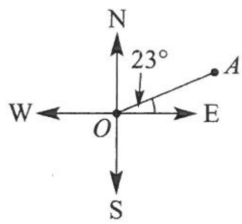
a


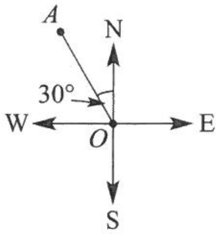
b


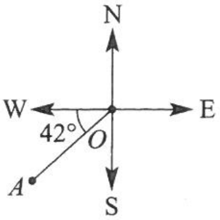
c


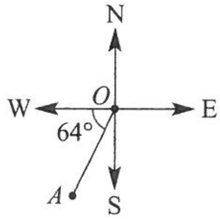
d


e


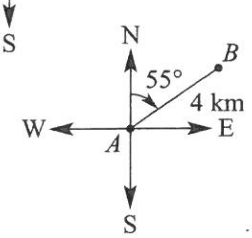
f


g


h


i


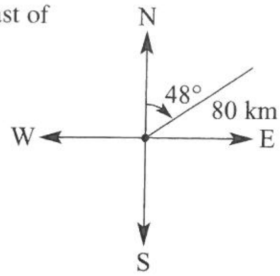
Example 19 4 A bushwalker walks 4 km on a true bearing of 055° from point *A* to point *B*. Find how far east point *B* is from point *A*, correct to two decimal places.



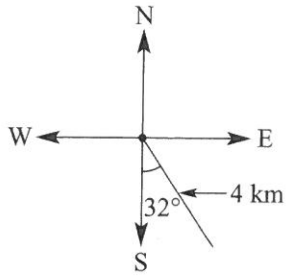
Ex 3J

Fluency

5 A speed boat travels 80 km on a true bearing of 048°. Find how far east of its starting point the speed boat is, correct to two decimal places.



6 After walking due east, then turning and walking due south, a hiker is 4 km 148° T from her starting point. Find how far she walked in a southerly direction, correct to one decimal place.



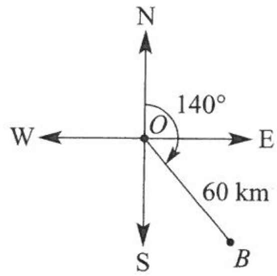
7 A four-wheel drive vehicle travels for 32 km on a true bearing of 200°. How far west (to the nearest km) of its starting point is it?

Problem-solving

Example 20 8 A fishing boat starts from point O and sails 60 km on a true bearing of 140° to point B.

a How far east of its starting point is the boat, to the nearest kilometre?

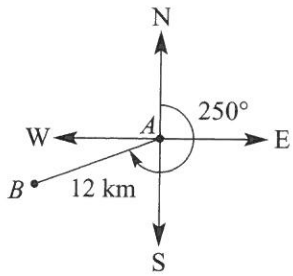
b What is the bearing of O from B?



9 Two towns, A and B, are 12 km apart. The true bearing of B from A is 250°.

a How far west of A is B, correct to one decimal place?

b Find the bearing of A from B.



10 A helicopter flies on a bearing of 140° for 210 km then flies due east for 175 km. How far east (to the nearest kilometre) has the helicopter travelled from its starting point?

11 Christopher walks 5 km south then walks on a true bearing of 036° until he is due east of his starting point. How far is he from his starting point, to one decimal place?

12 Two cyclists leave from the same starting point. One cyclist travels due west while the other travels on a bearing of 202°. After travelling for 18 km, the second cyclist is due south of the first cyclist. How far (to the nearest metre) has the first cyclist travelled?

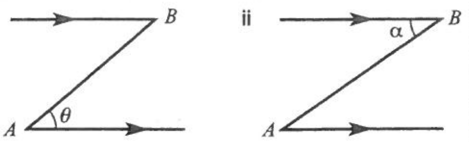
ANSWERS

Exercise 3I

1 $a = 65, b = 25$

2 a 22° **b** 22°

3 a i ii



b yes, $\theta = \alpha$, alternate angles are equal on parallel lines

4 29 m **5** 16 m **6** 157 m **7** 38 m

8 90 m **9** 37° **10** 6° **11** 10°

12 yes by 244.8 m **13 a** 6° **b** 209 m

14 4634 mm **15** 15°, 4319 mm **16** 1.25 m

17 a 6.86 m **b** 26.6° **c i** $m = h + y$

ii $y = x \tan \theta$ **iii** $m = h + x \tan\theta$

18 $A = \frac{1}{2} a^2 \tan \theta$

19 a i 47.5 km **ii** 16.25 km **b** no, $\sin(2 \times \theta) < 2 \times \sin \theta$

c yes, $\sin\left(\frac{1}{2}\theta\right) > \frac{1}{2}\sin\theta$

20 a yes **b** 12.42 km

c no, after 10 mins will be above 4 km

d 93 km/h or more

21 a 1312 m **b** 236.16 km/h

Exercise 3J

1 a 0° **b** 045° **c** 090° **d** 135°

e 180° **f** 225° **g** 270° **h** 315°

2 a 070° **b** 130° **c** 255° **d** 332°

3 a i 040°T **ii** 220°T **b i** 142°T **ii** 322°T

c i 210°T **ii** 030°T **d i** 288°T **ii** 108°T

e i 125°T **ii** 305°T **f i** 067°T **ii** 247°T

g i 330°T **ii** 150°T **h i** 228°T **ii** 048°T

i i 206°T **ii** 26°T

4 3.28 km **5** 59.45 km **6** 3.4 km **7** 11 km

8 a 39 km **b** 320°T **9 a** 11.3 km **b** 070°T

10 310 km **11** 3.6 km **12** 6.743 km

13 a $180 + a$ **b** $a - 180$

14 a 320° **b** 245° **c** 065° **d** 238° **e** 278°

15 a 620 km **b** 606 km **c** 129 km

16 a i 115 km **ii** 96 km **b** 158 km

c i 68 km **ii** 39.5 min

Investigation